Drive Easy

A full-stack Spring Boot application for real-time vehicle browsing, hourly booking, and fleet management.

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Introduction

Problem Statement

Traditional vehicle rental relies on manual paperwork, phone calls, and limited availability visibility. Users struggle to find, book, and manage rentals in real time.

Objective

Build a full-stack web platform where users can browse, search by date/time, and book vehicles online, while admins manage the fleet and booking lifecycle.

Scope

User registration JWT auth, vehicle catalog with filtering, hourly booking with conflict detection, booking lifecycle (PENDING → ACTIVE → COMPLETED/CANCELLED), and an admin dashboard.

Key Highlights

- JWT-based authentication
- Role-based access (User / Admin)
- Hourly booking with conflict detection
- 30-day maximum rental limit
- Soft-delete fleet management
- Paginated REST APIs
- Booking lifecycle tracking

Tech Stack

Backend

- Java 17 — Language
- Spring Boot 3 — Framework
- Spring Security — Auth & JWT
- Spring Data JPA — ORM
- PostgreSQL — Database
- Maven — Build Tool

Frontend

- HTML5 — Structure
- CSS3 — Styling
- JavaScript — Interactivity

Dev & Tools

- IntelliJ IDEA — IDE

Features & Functionality

Guest

- Register & Login (JWT)
- Browse vehicle catalog
- Filter by type & category
- Search by date & time
- View vehicle details

User

- Create PENDING booking
- Confirm booking (→ ACTIVE)
- Cancel booking
- View own booking history
- Update profile & password

Admin

- Add / Edit vehicles
- Toggle vehicle status
- Manage categories
- View all bookings

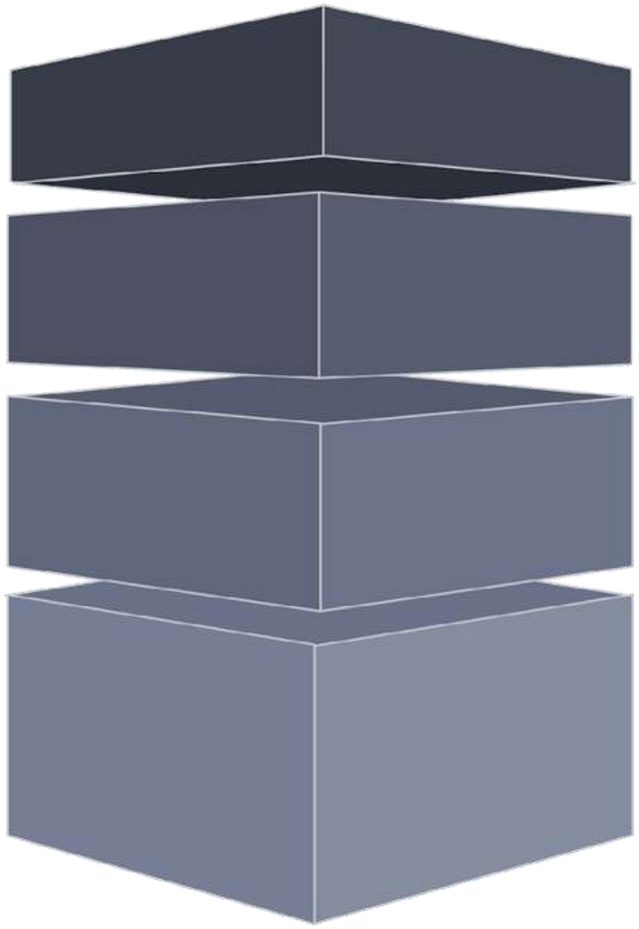
Architecture Diagram

Client Layer

HTML, CSS, JS making REST calls.

Application Layer

Spring Boot 3: Controller, Service, Repository.



API Gateway

JWT auth filter with Spring Security.

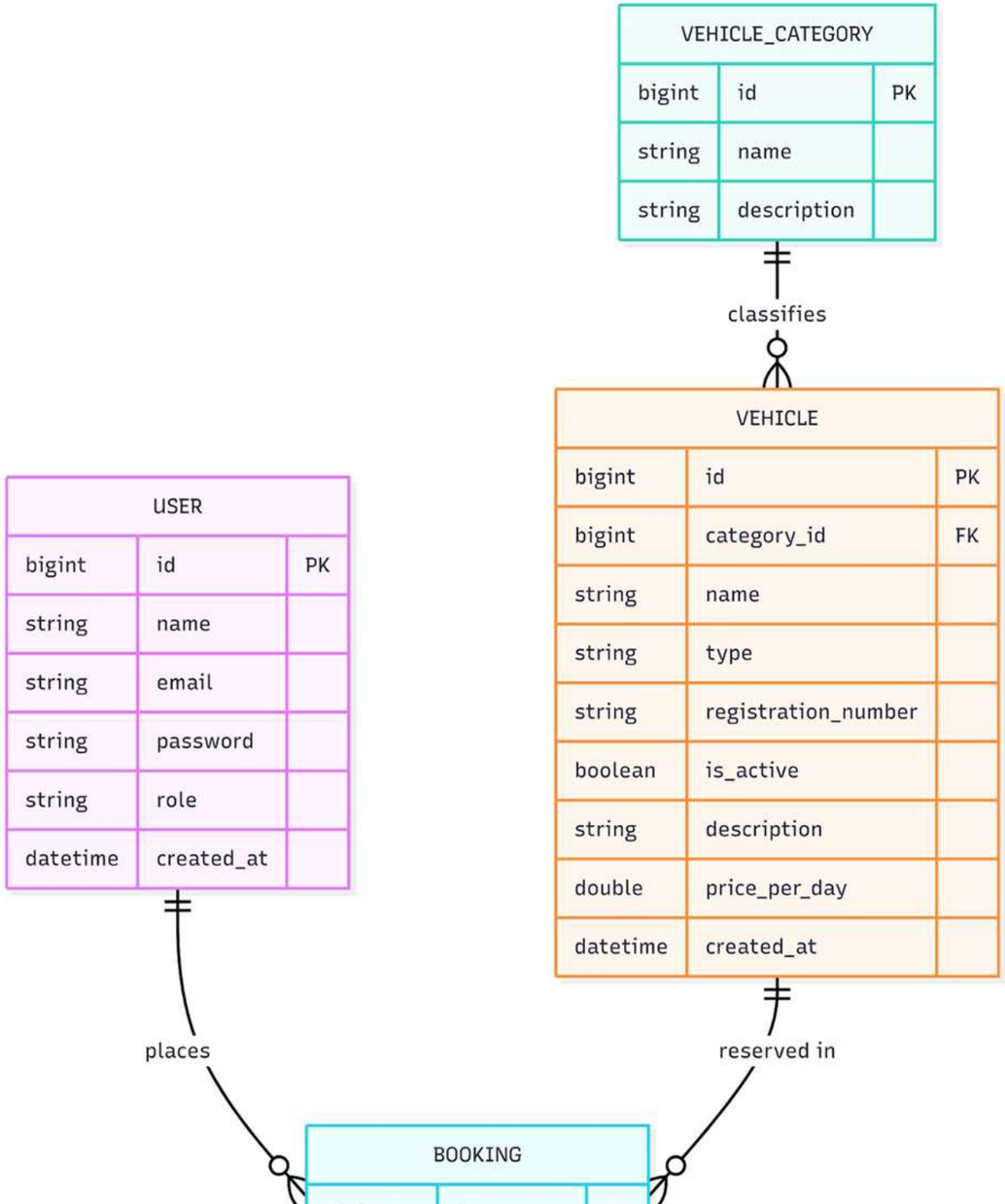
Database Layer

PostgreSQL storing users, vehicles, bookings, categories.

The system follows a clean layered architecture: the JS frontend communicates via REST with a Spring Boot backend secured by JWT tokens, backed by PostgreSQL for persistent data storage.

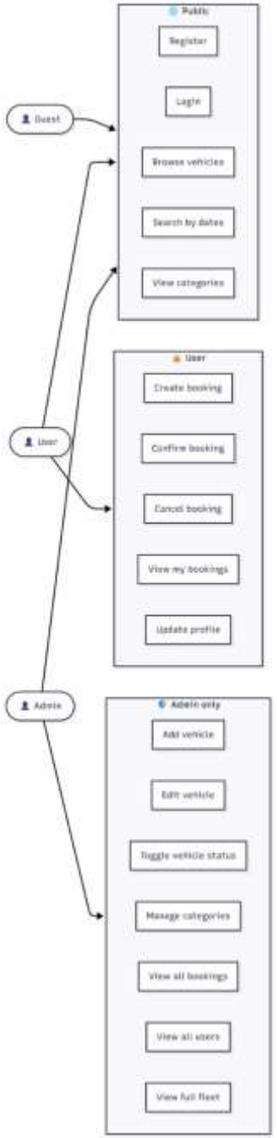
Entity-Relationship Diagram

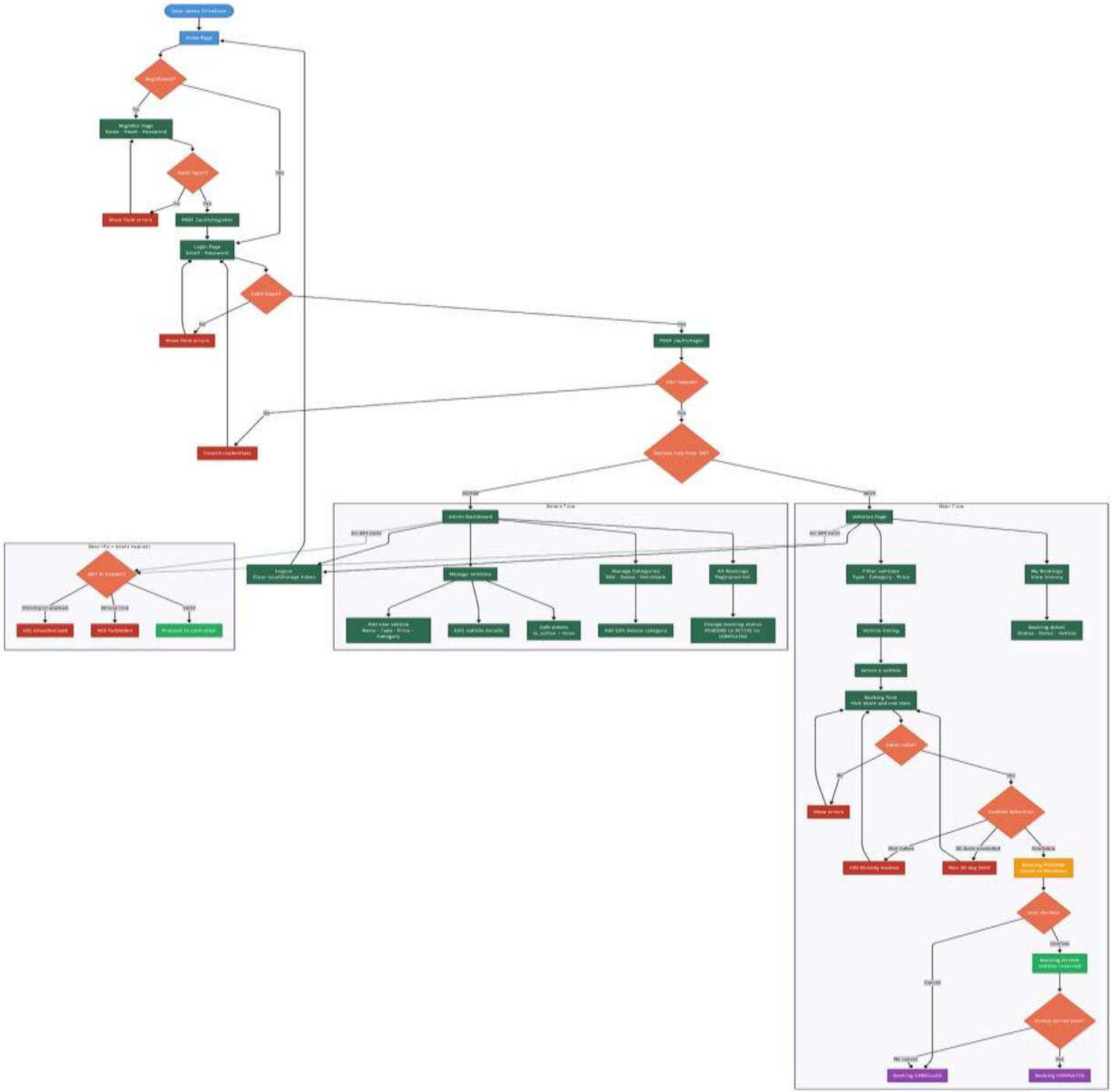
Core Entities



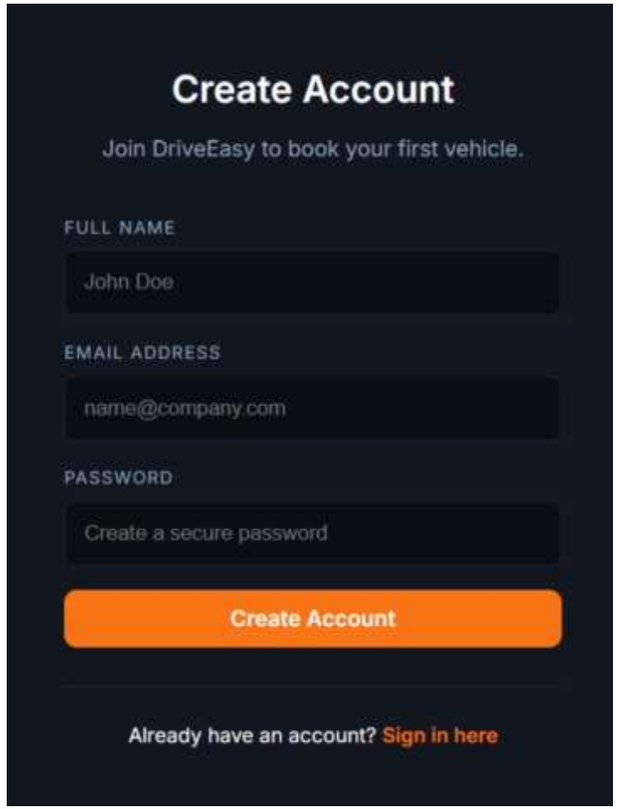
Use Case Diagram

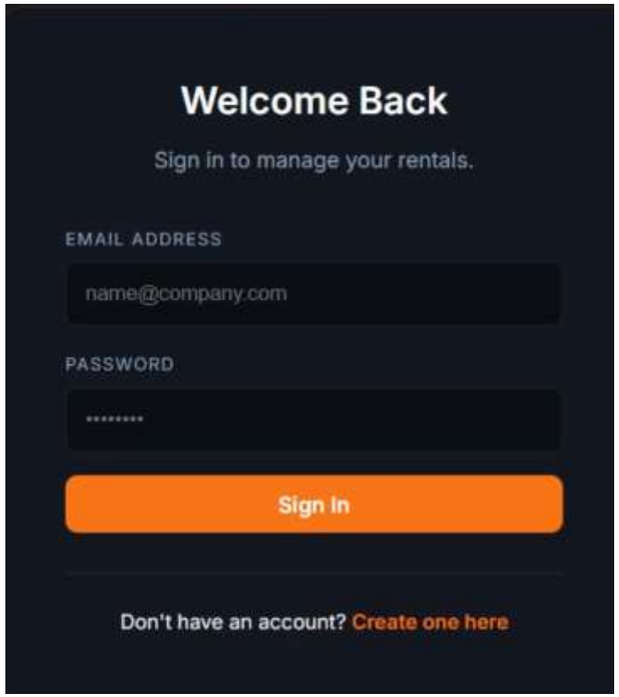
Actors & Use Cases

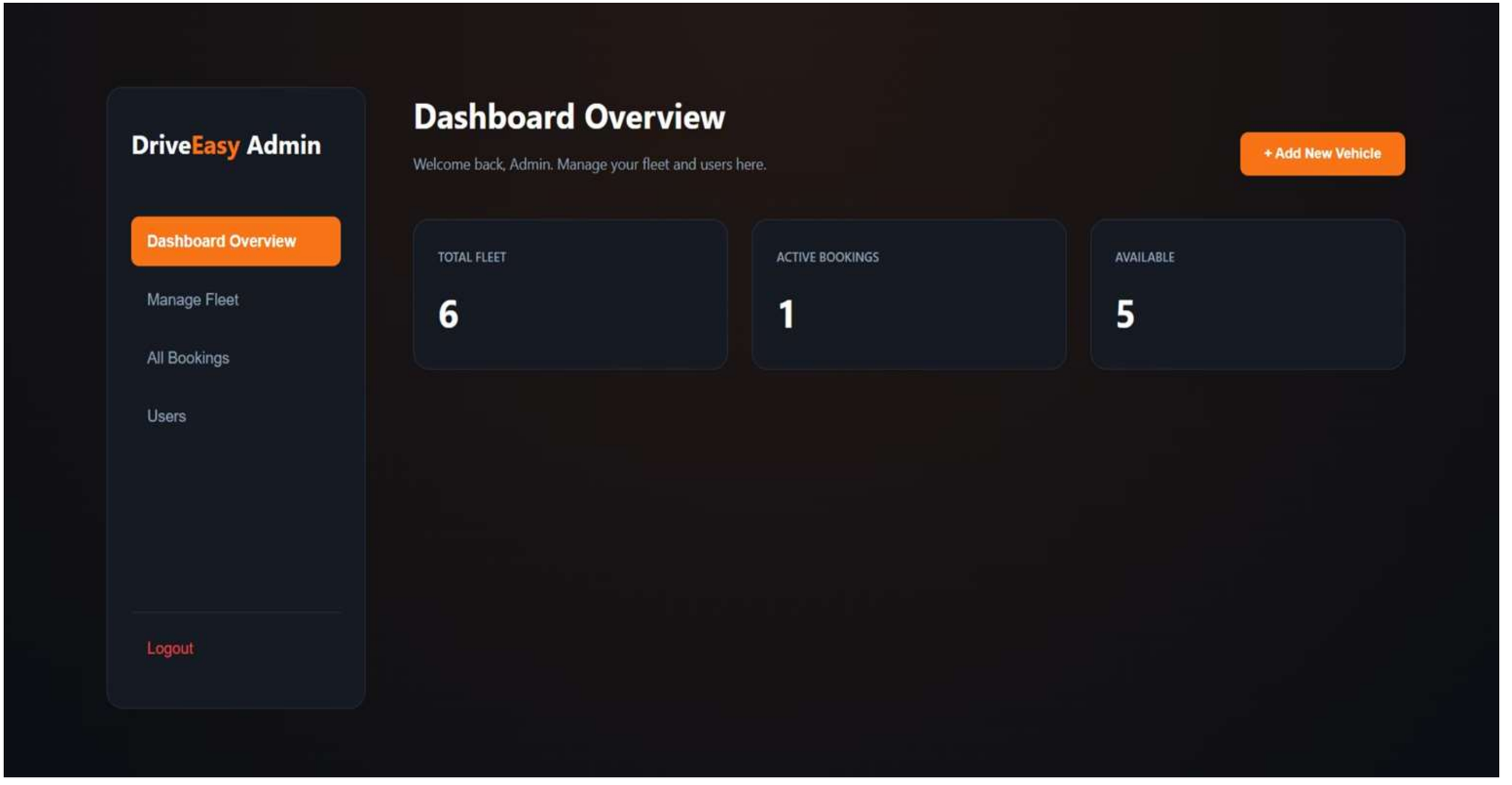




Application






DriveEasy Admin
Dashboard Overview
Manage Fleet
All Bookings
Users
Logout
Dashboard Overview
Welcome back, Admin. Manage your fleet and users here.
+ Add New Vehicle
TOTAL FLEET
6
ACTIVE BOOKINGS
1
AVAILABLE
5

My Bookings

View and manage your current and past rentals.

Seltos
Booking ID: #BK-6
02-05-2026 18:58 → 08-05-2026 15:58

ACTIVE
₹33,600
Locked (Trip Started)
The rental period has al

mahindra thar
Booking ID: #BK-5
30-04-2026 13:47 → 30-04-2026 13:48

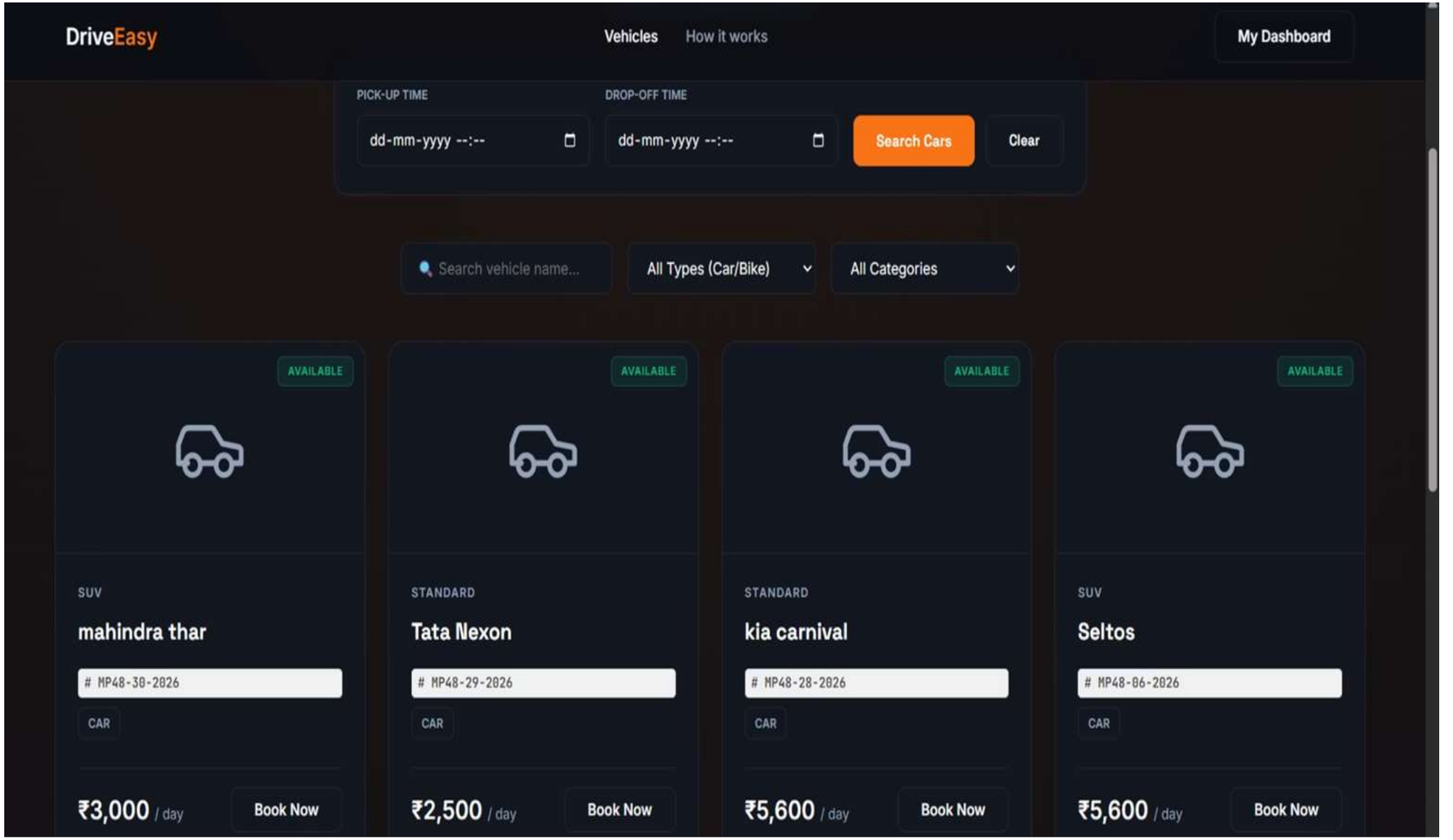
COMPLETED
₹3,000

mahindra thar
Booking ID: #BK-4
30-04-2026 13:36 → 30-04-2026 13:36

COMPLETED
₹3,000

Tata Nexon
Booking ID: #BK-3
30-04-2026 16:41 → 30-04-2026 16:49

COMPLETED
₹2,500


DriveEasy
Vehicles
How it works
My Dashboard
PICK-UP TIME
dd-mm-yyyy --:--
DROP-OFF TIME
dd-mm-yyyy --:--
Search Cars
Clear
Search vehicle name...
All Types (Car/Bike)
All Categories
AVAILABLE
SUV
mahindra thar
MP48-30-2026
CAR
₹3,000 / day
Book Now
AVAILABLE
STANDARD
Tata Nexon
MP48-29-2026
CAR
₹2,500 / day
Book Now
AVAILABLE
STANDARD
kia carnival
MP48-28-2026
CAR
₹5,600 / day
Book Now
AVAILABLE
SUV
Seltos
MP48-06-2026
CAR
₹5,600 / day
Book Now

Thank You

Drive Easy — Minor CapStone Project 2026

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